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FIT3080: Artificial Intelligence

Bayesian Networks II: d-separation, Inference, training & evaluation, decision networks

Lecturer: Steven Mascaro

Readings: Russell & Norvig 16.1-2, 16.5

Korb & Nicholson, 2011: 3.1-3.6, 4.1-4.4, Ch 10 plus papers (see end)

Slides are adapted from those of Ingrid Zukerman, Ann Nicholson, Stuart Russell, Andrew Moore, Dan Klein, Kevin Korb and Yue Wu

Bayesian Reasoning I: Learning Objectives

Learning objectives:

- Understand d-separation
- Understand basics of BN inference
- Perform reasoning in Netica
- Learn BN parameters from data
- Perform basic evaluation of BN models with data
- Build and use decision networks

d-Separation (Directional Separation)

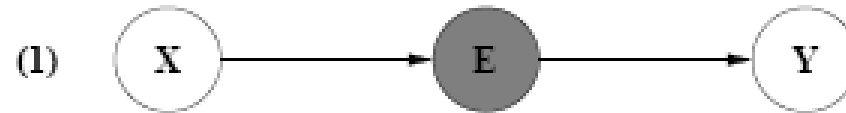
- Lets us work out conditional independencies just from the graph
- e.g. In the following, we know X and Y are independent given evidence on E , without checking the probabilities



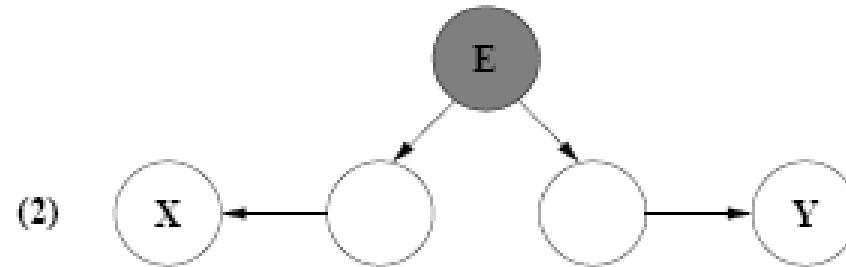
- Recall conditional independence here would mean:
 $\Pr(X|Y \wedge E) = \Pr(X|E)$ and also $\Pr(Y|X \wedge E) = \Pr(Y|E)$
- We can write this as $X \perp\!\!\!\perp Y | E$
- d-separation works if Markov property holds (generally assumed)

Determining d-separation

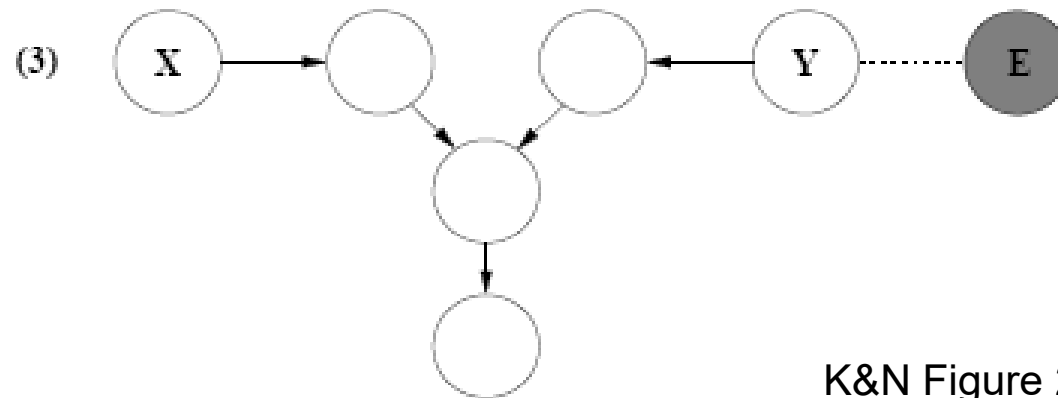
Chain



Common
cause



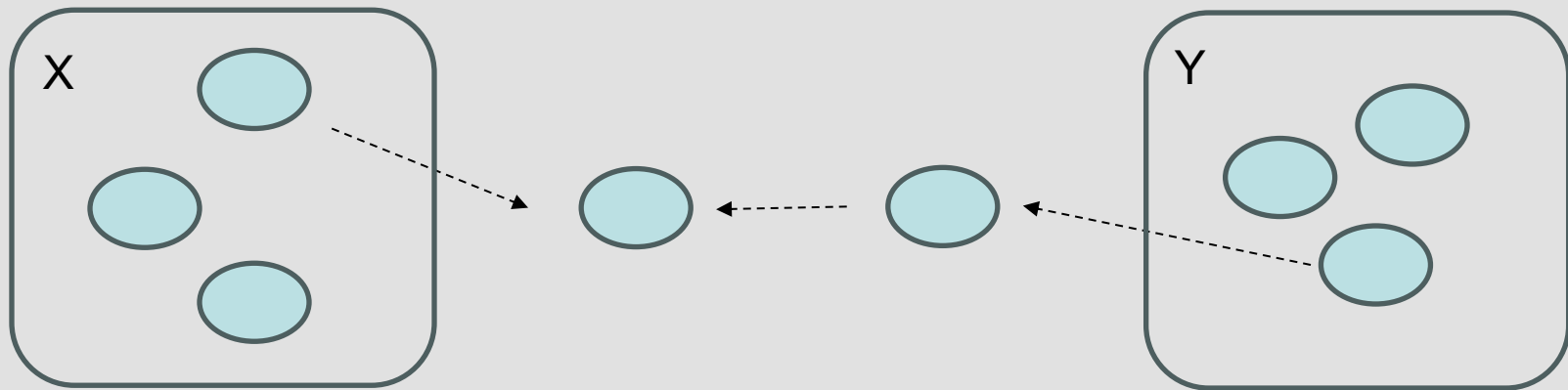
Common
effect



K&N Figure 2.5

d-separation – Path

Undirected path: A path between two sets of nodes X and Y is any sequence of nodes between a member of X and a member of Y such that every adjacent pair of nodes is connected by an arc (regardless of direction), and no node appears in the sequence twice

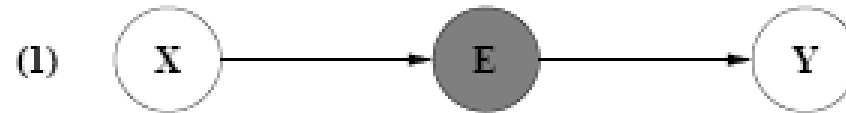


d-separation – Blocked Path

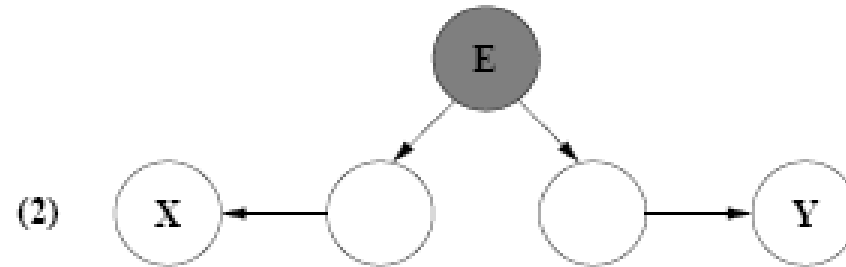
- **Blocked path:** A path is **blocked** given a set of nodes **E**, if there is a node **Z** on the path for which at least one of three conditions holds:
 1. **Chain:** **Z** is in **E** and **Z** has one arrow on the path leading in and one arrow out
 2. **Common cause:** **Z** is in **E** and **Z** has both path arrows leading out
 3. **Common effect:** Neither **Z** nor any descendant of **Z** is in **E**, and both path arrows lead into **Z**
- A set of nodes **E** **d-separates** two sets of nodes **X** and **Y**, if every undirected path from a node in **X** to a node in **Y** is **blocked** given **E**

Determining d-separation

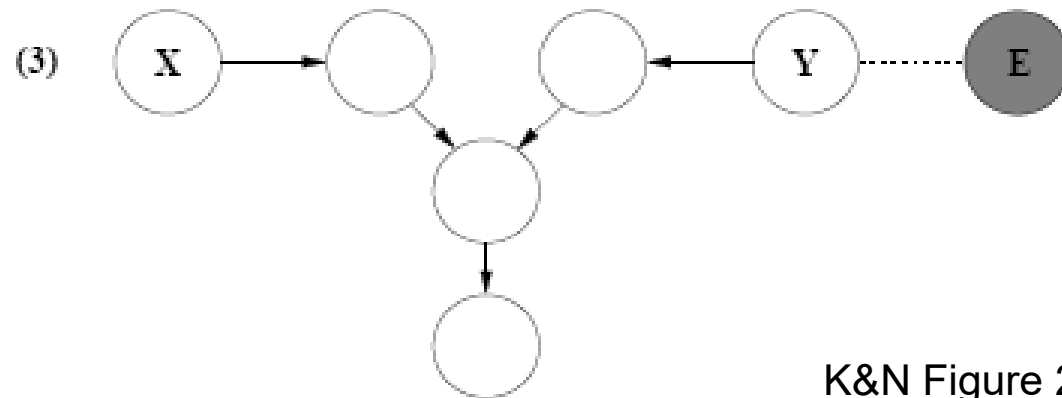
Chain



Common
cause



Common
effect



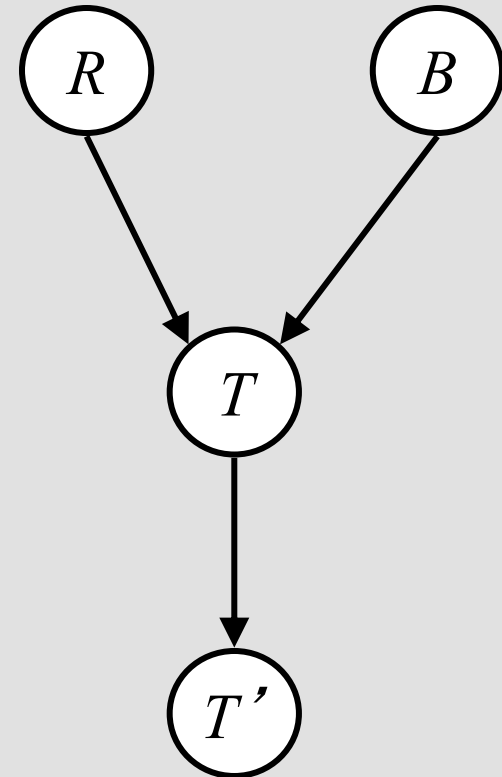
K&N Figure 2.5

d-separation – Example (I)

$$R \perp\!\!\!\perp B$$

$$R \perp\!\!\!\perp B | T$$

$$R \perp\!\!\!\perp B | T'$$



d-separation – Example (II)

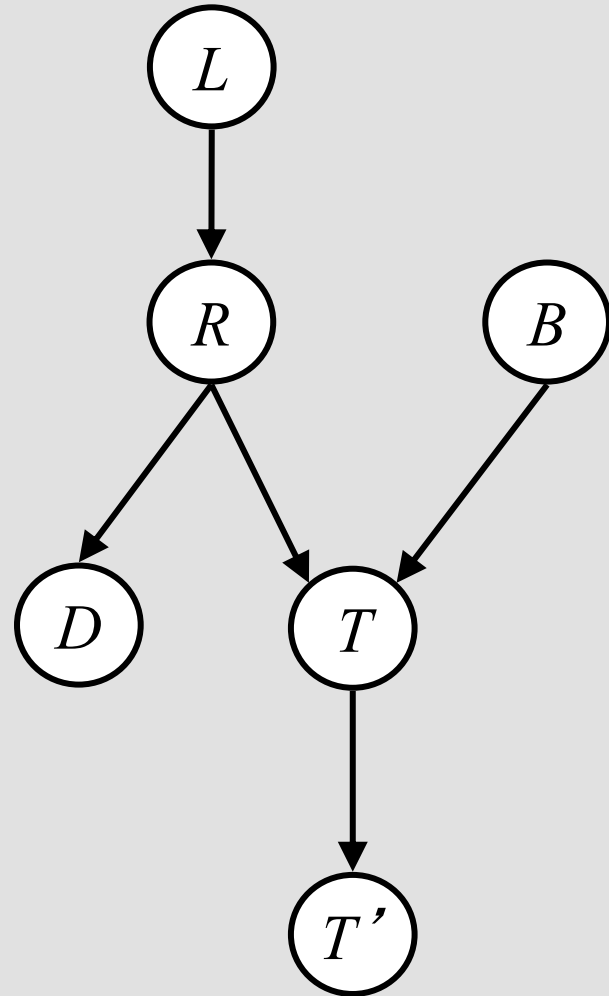
$$L \perp\!\!\!\perp T' | T$$

$$L \perp\!\!\!\perp B$$

$$L \perp\!\!\!\perp B | T$$

$$L \perp\!\!\!\perp B | T'$$

$$L \perp\!\!\!\perp B | T, R$$



d-separation – Example (III)

- **Variables:**

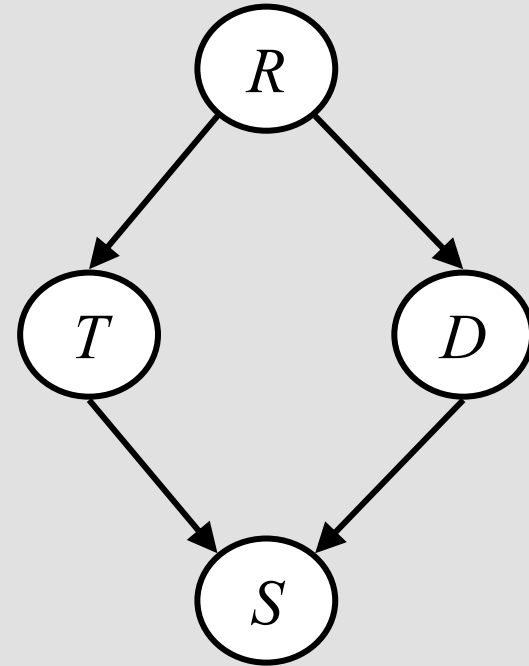
- R: Raining
- T: Traffic
- D: Roof drips
- S: I'm sad

- **Questions:**

$$T \perp\!\!\!\perp D$$

$$T \perp\!\!\!\perp D | R$$

$$T \perp\!\!\!\perp D | R, S$$





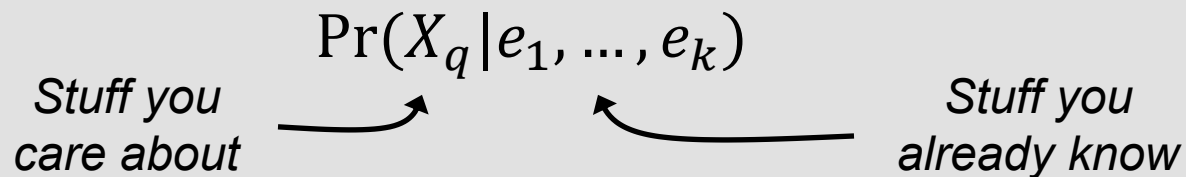
FIT3080 – Artificial Intelligence

Inference in Bayesian Networks

Korb & Nicholson, 2011: Chapter 3

RECAP: BN inference task

- Basic task for any probabilistic inference system:
Compute the posterior probability distribution for a set of *query variables*, given new information about some *evidence variables*
- Also called *conditioning* or *belief updating* or *inference*



Inference in BNs: Overview

- **Exact inference**

- Enumeration
- Variable elimination
- Message passing
- Junction-tree algorithms (Netica “compilation”)

All general inference approaches are NP hard!

- **Approximate inference**

- Stochastic simulation
 - > Logic sampling
 - > Likelihood weighting
 - > MCMC (Markov Chain Monte Carlo)/Gibbs sampling

Including exact and stochastic (to specified precision ϵ)

- **Uncertain (“virtual”) evidence**

- Can use same techniques

But many special and sparse cases feasible

- **Most probable explanation (MPE)**

- $\underset{q}{\operatorname{argmax}} \Pr(Q = q | E_1 = e_1, \dots, E_k = e_k) \quad \forall Q \notin E$
- Special case of MAP (maximum a posteriori probability)

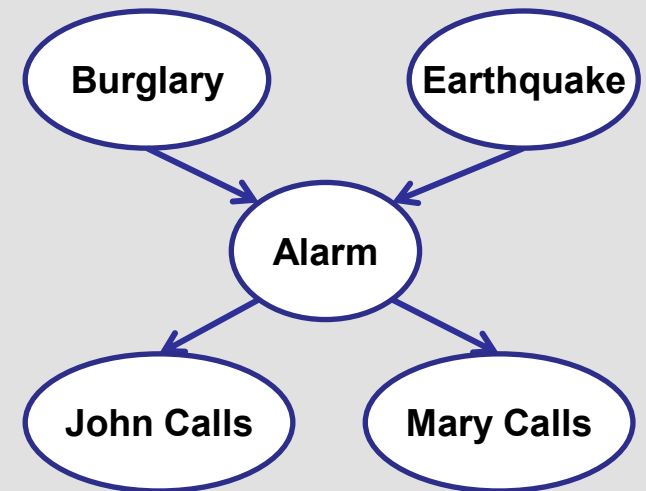


Pearl's Burglar Alarm BN

Your new burglar alarm reliably detects burglary, but also responds to minor earthquakes.

Two neighbors, John and Mary, promise to call you when they hear the alarm. John always calls when he hears the alarm, but can confuse the alarm with the phone ringing and calls then also. On the other hand, Mary likes loud music and sometimes doesn't hear the alarm.

Given evidence about who has and hasn't called, you'd like to estimate the probability of a burglary. (From Pearl (1988).)

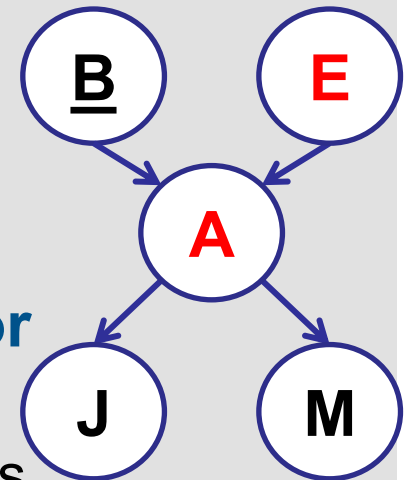


Inference by Enumeration

- **Given unlimited time, inference in BNs is easy**
- **Recipe:**
 - State the **marginal** probabilities you need
 - Figure out ALL the **joint** probabilities you need
 - Calculate and combine them
- **Example:**

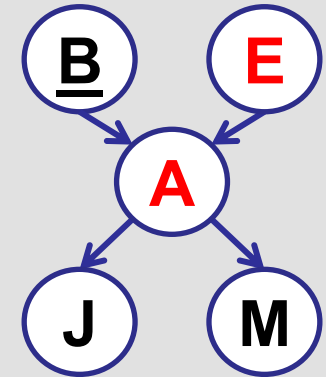
$$\Pr(+b \mid +j, +m) = \frac{\Pr(+b, +j, +m)}{\Pr(+j, +m)}$$

- **There is no need to calculate the denominator**
 - $\Pr(\neg b \mid +j, +m)$ has the same denominator
 - Replace the denominator with α in both equations
 - $\Pr(+b \mid +j, +m) + \Pr(\neg b \mid +j, +m) =$
 $\alpha \Pr(+b, +j, +m) + \alpha \Pr(\neg b, +j, +m) = 1$



Enumeration – Example (I)

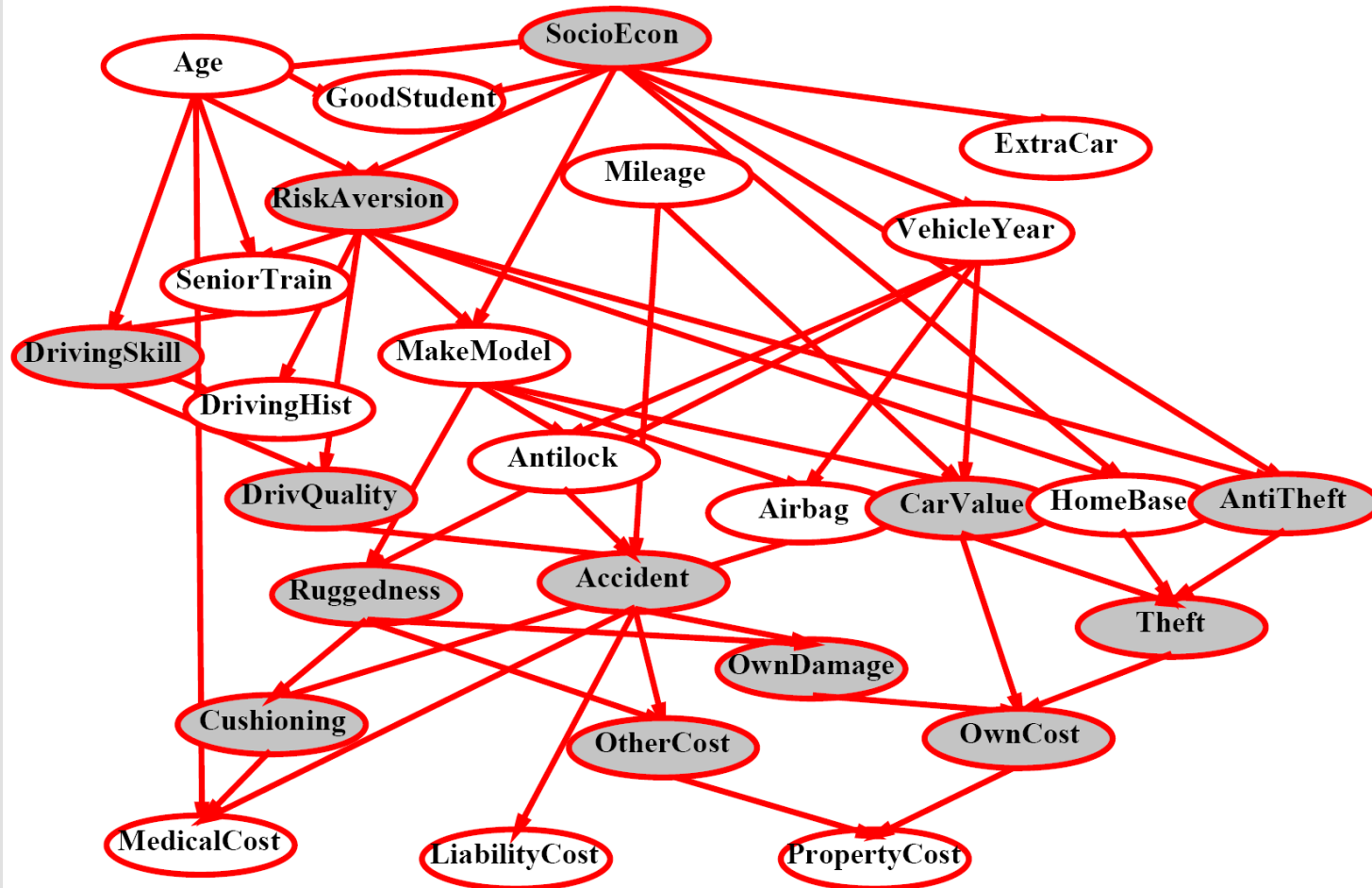
- In this simple method, we only need the BN to synthesize the needed joint distribution entries from the CPTs



$$\Pr(+b, +j, +m) =$$

$$\begin{aligned} & \Pr(+b) \Pr(+e) \Pr(+a|+b, +e) \Pr(+j|+a) \Pr(+m|+a) + \\ & \Pr(+b) \Pr(+e) \Pr(-a|+b, +e) \Pr(+j|-a) \Pr(+m|-a) + \\ & \Pr(+b) \Pr(-e) \Pr(+a|+b, -e) \Pr(+j|+a) \Pr(+m|+a) + \\ & \Pr(+b) \Pr(-e) \Pr(-a|+b, -e) \Pr(+j|-a) \Pr(+m|-a) \end{aligned}$$

Inference by Enumeration – Practical?



Variable Elimination (VE)

- **Why is inference by enumeration so slow?**
 - You join up the whole joint distribution before you sum out the hidden variables
 - You end up repeating a lot of work!
- **Idea: do the calculation once and save the results for later use**
 - Called “Variable Elimination”
 - Interleave joining and marginalizing
 - Still NP-hard (all methods are NP hard in general)
 - But usually much faster than inference by enumeration

VE: Simple example

Our query: $\Pr(B \mid +j, +m)$

Choose elimination order: E, A

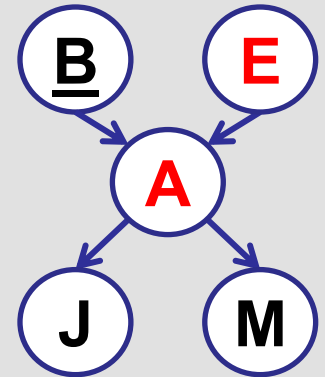
(any order works, but some produce smaller intermediate products)

Eliminate E:
$$\sum_E \Pr(E) \Pr(A \mid B, E) = \Pr(A \mid B)$$

Eliminate A:
$$\sum_A \Pr(+j \mid A) \Pr(+m \mid A) \Pr(A \mid B) = \Pr(+j, +m \mid B)$$

Combine:
$$\Pr(B) \Pr(+j, +m \mid B) = \Pr(B, +j, +m)$$

Normalise:
$$\frac{\Pr(B, +j, +m)}{\sum_B \Pr(B, +j, +m)} = \Pr(B \mid +j, +m)$$



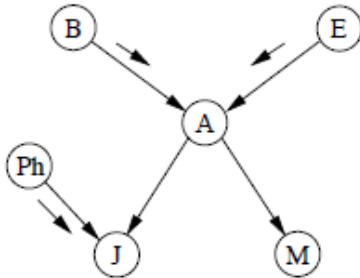
Improving efficiency

- **VE is good, but:**
 1. We still throw away all calculations after each query
 2. We need some way to choose a good elimination order
 - **Message passing solves (1)**
 - Only pass around update messages:
 - > with no evidence (update everything)
 - > when new evidence observed (update products affected by new evidence)
 - Can only be used with polytrees (at least, with predictable results)
 - **Clustering/junction trees solve (2)**
 - NP hard, but sparse graphs can efficiently be converted to trees
 - We can then use message passing
 - Used in most BN packages like Netica, GeNIe, pgmpy, etc.
 - (Not covered in this unit)
- Note: there are other ways to do both

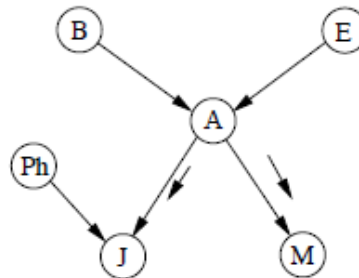
Message passing

PROPAGATION, NO EVIDENCE

PHASE 1

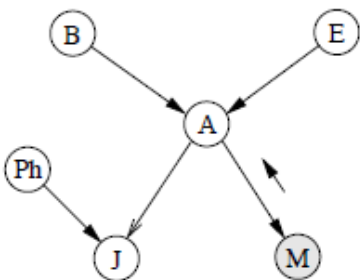


PHASE 2

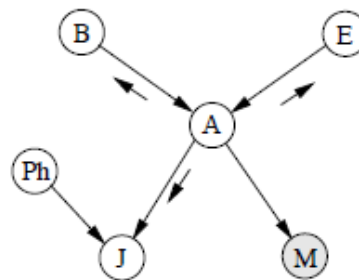


PROPAGATION, EVIDENCE for node M

PHASE 1



PHASE 2



PHASE 3

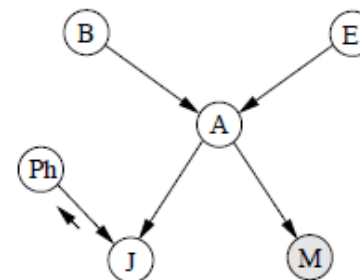


FIGURE 3.4: Message passing algorithm propagation stages for without evidence (above) and with evidence for node *M* (below).

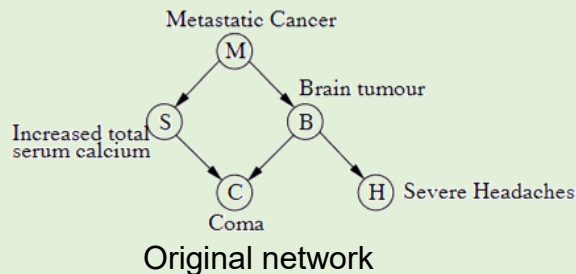
Message passing

- All computations are local $\rightarrow$ efficient (linear in number of variables, in fact)
- Requires summation over all joint instantiations of parent nodes $\rightarrow$ exponential in no. of parents.
- No. of propagation steps depends on length of longest path
- Trees/polytrees only

Clustering

Remove loops by merging nodes, giving a polytree

Example (Transformed Metastatic Cancer)

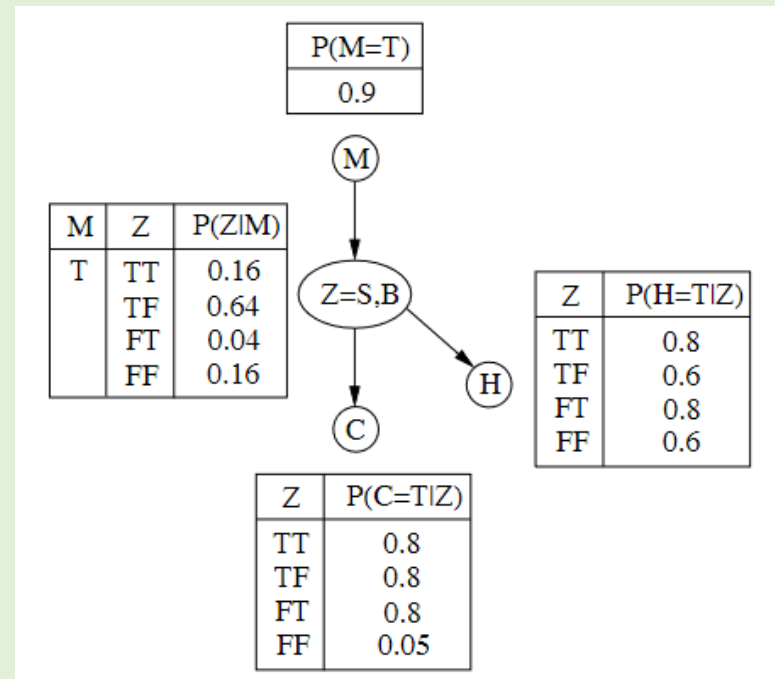


New node Z combines S & B

$$P(Z|M) = P(S, B|M) = P(S|M)P(B|M)$$

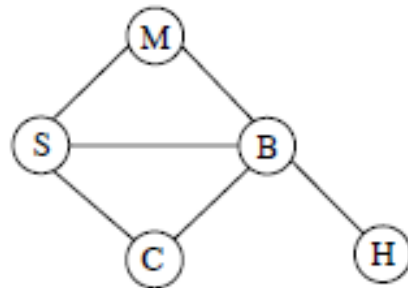
$$P(H|Z) = P(H|S, B) = P(H|B)$$

$$P(C|Z) = P(C|S, B)$$

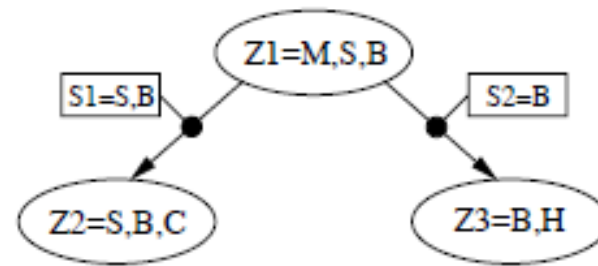


Junction trees

- Junction trees most efficient algorithm in this class (e.g. used in Hugin (Jensen, 1996), Netica).



(a)



(b)

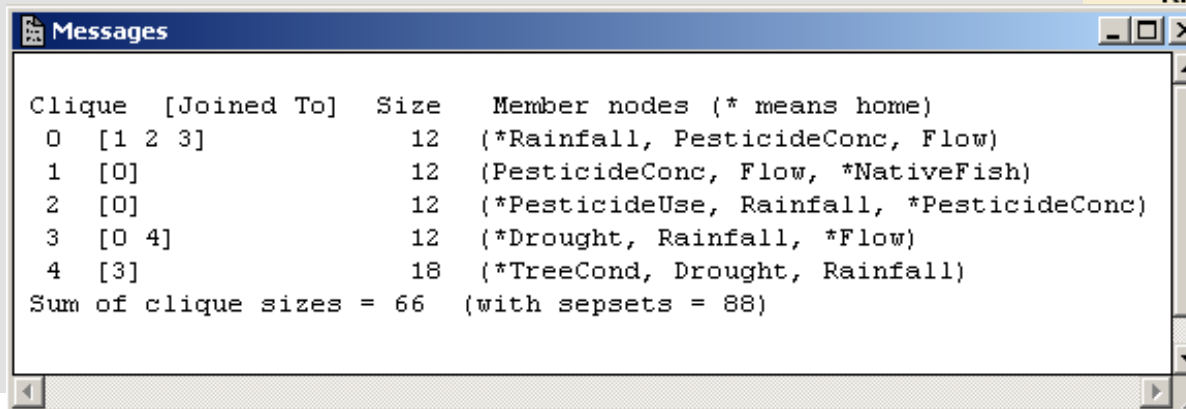
(a) Converted to a triangulated, undirected graph

(b) Polytree with new “supernodes”

Junction trees in Netica

Need 2 steps:

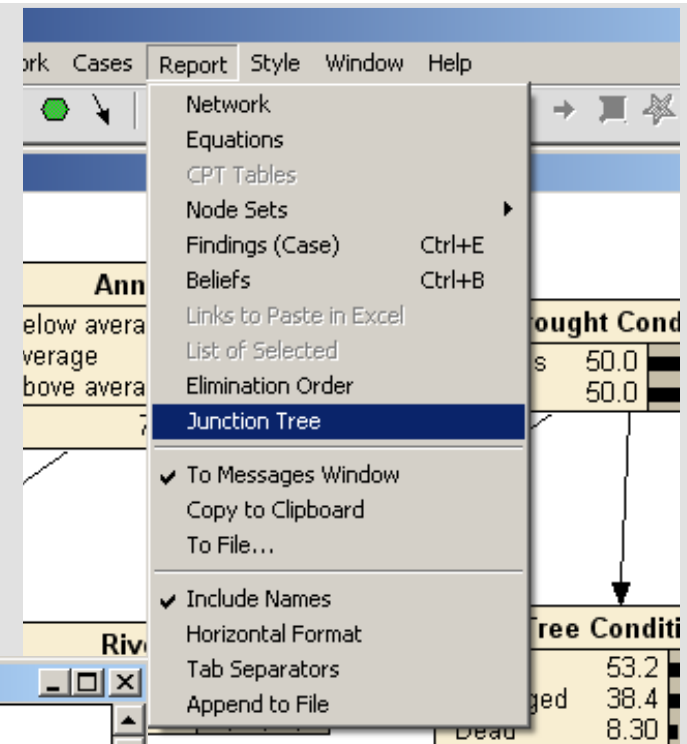
1. **Compile** into junction tree
(can be slow)
2. Do belief updating/message passing
(usually fast)



The screenshot shows the 'Messages' window in Netica, displaying the results of a compilation process. It lists five cliques with their members and sizes, and provides a summary of the total clique sizes and sepsets.

Clique	[Joined To]	Size	Member nodes (* means home)
0	[1 2 3]	12	(*Rainfall, PesticideConc, Flow)
1	[0]	12	(PesticideConc, Flow, *NativeFish)
2	[0]	12	(*PesticideUse, Rainfall, *PesticideConc)
3	[0 4]	12	(*Drought, Rainfall, *Flow)
4	[3]	18	(*TreeCond, Drought, Rainfall)

Sum of clique sizes = 66 (with sepsets = 88)





FIT3080 – Artificial Intelligence

Reasoning, Learning Probabilities from Data & Evaluation

Korb & Nicholson, 2011: Ch 2.3, 6.2, 7.5

Reasoning: Types of Evidence

(Called "Findings" in Netica)

Specific evidence:

- A definite finding that a node X has a particular value, x .
- Example: *PesticideUse = High*

Negative evidence:

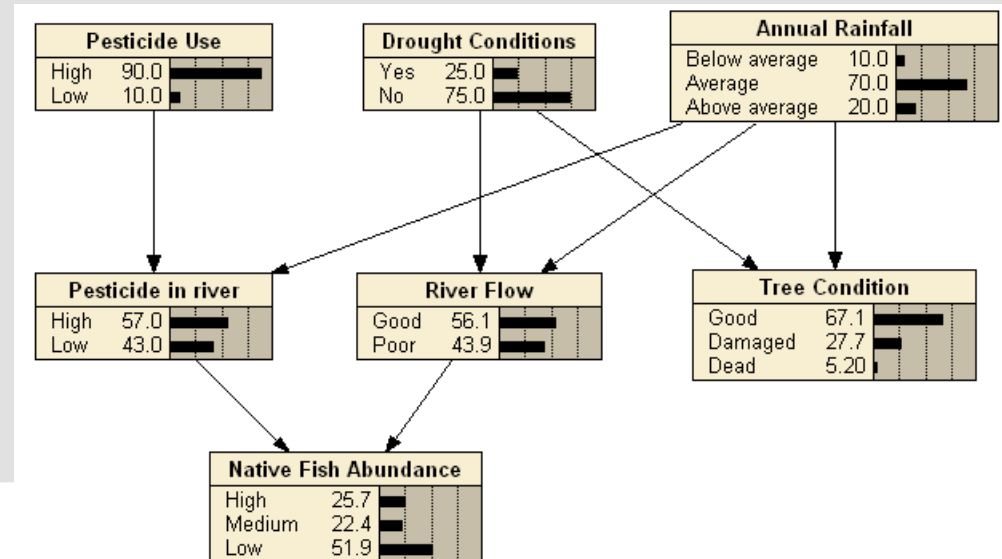
- A finding that node Y is not in state y_1 (but may take any other values)
- Example: *TreeCondition $\neq$ Dead*



Types of evidence

Uncertain evidence:

- Source of information is not perfect reflection of the true state of the world.
- Example: A test that signals high pesticide 80% of the time when *Pesticide in River=High*, 10% of the time when *Low*.



Using a “virtual node”

- One option is to add a new “virtual” node Observation, with states *High* and *Low* (lab exercise)

The diagram shows a Bayesian network with two nodes: "Pesticide in river" and "Observation".

Pesticide in river

High	91.4
Low	8.62

Observation

High	100
Low	0

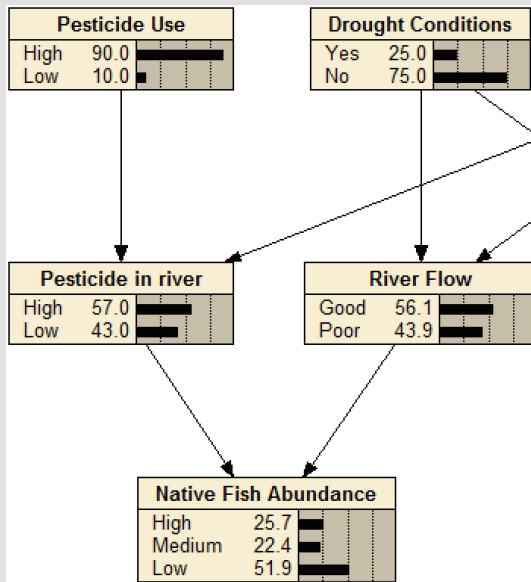
Node: **Observation** [Apply] [Okay]

[Chance] [% Probability] [Reset] [Close]

Pesticide in river	High	Low
High	80	20
Low	10	90

Using Netica's "likelihood evidence"

- Another option is to use Netica's likelihood evidence feature



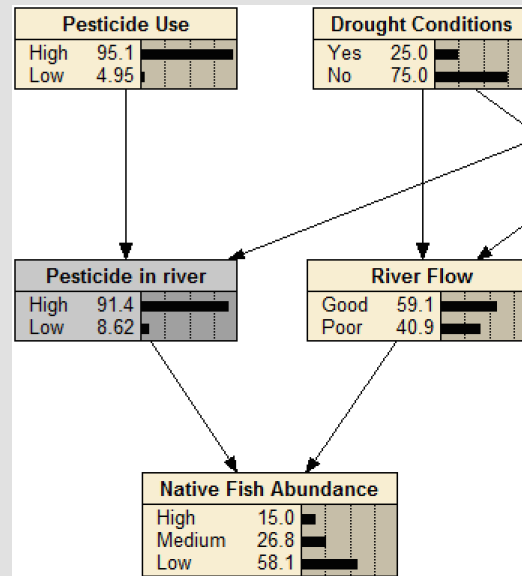
(no evidence)

Enter P[Observation | PesticideInRiver=High]

Okay Revert Cancel

Enter P[Observation | PesticideInRiver=Low]

Okay Revert Cancel



{0.8,0.1}
likelihood
evidence

Types of Evidence

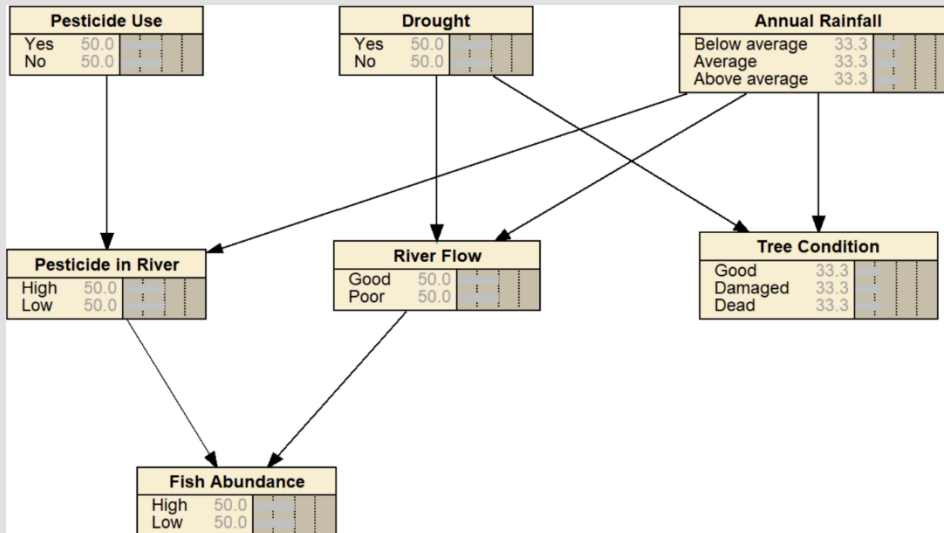
**Netica Demo
(See lecture video)**



Learning Probabilities from Data

- **We can obtain parameter estimates from:**
 - Our own estimates
 - Experts
 - Literature & other models
 - Data
- **Multiple ways to learn from data**
- **Common approaches:**
 - Count learning
 - Expectation-Maximisation (EM)

Count Learning



1. Initialise the count for each CPT cell (to some prior)
2. Step through each row of the data
3. Increment a CPT cell count each time we see its value occur together with the values of its parent nodes
4. Normalise the counts in each row into a probability distribution

Count Learning – Example

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average		
Yes	Average		
Yes	Above Average		
No	Below Average		
No	Average		
No	Above Average		

Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
Average	Damaged	Yes	High	Good	No	Low
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning – Example

Start with a prior of 1 for each cell

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average	1	1
Yes	Average	1	1
Yes	Above Average	1	1
No	Below Average	1	1
No	Average	1	1
No	Above Average	1	1

Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
Average	Damaged	Yes	High	Good	No	Low
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning – Example

For each case data row, increment related cell count in CPT

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average	1	1
Yes	Average	1	1
Yes	Above Average	1	1
No	Below Average	1	1
No	Average	1	2
No	Above Average	1	1



Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
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Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning – Example

For each case data row, increment related cell count in CPT

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average	1	1
Yes	Average	1	1
Yes	Above Average	1	1
No	Below Average	1	1
No	Average	1	3
No	Above Average	1	1



Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
Average	Damaged	Yes	High	Good	No	Low
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning – Example

For each case data row, increment related cell count in CPT

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average	1	1
Yes	Average	1	1
Yes	Above Average	1	1
No	Below Average	1	1
No	Average	2	3
No	Above Average	1	1



Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
Average	Damaged	Yes	High	Good	No	Low
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning – Example

For each case data row, increment related cell count in CPT

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average	1	2
Yes	Average	3	9
Yes	Above Average	4	3
No	Below Average	5	3
No	Average	37	29
No	Above Average	16	1

Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
Average	Damaged	Yes	High	Good	No	Low
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning – Example

Finally, normalize to get probability values

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average	33%	67%
Yes	Average	25%	75%
Yes	Above Average	57%	43%
No	Below Average	57%	43%
No	Average	56%	44%
No	Above Average	94%	6%

Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	High
Above_average	Damaged	Yes	Low	Good	Yes	High
Average	Good	Yes	Low	Good	No	High
Average	Good	No	Low	Good	No	High
Average	Damaged	Yes	High	Good	No	Low
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	Low	Poor	No	Low
Average	Dead	Yes	Low	Poor	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	Low	Poor	No	Low
Average	Good	Yes	Low	Poor	No	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	Low	Good	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	High
Average	Good	Yes	High	Good	No	High
Average	Good	Yes	Low	Good	No	High
Above_average	Damaged	Yes	High	Good	No	Low
Average	Dead	Yes	High	Poor	Yes	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Damaged	No	Low	Poor	Yes	High
Average	Dead	Yes	High	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Above_average	Good	Yes	High	Good	No	Low
Average	Good	Yes	Low	Poor	No	Low



Count Learning

Missing data **will be discarded**

Drought	Rainfall	River Flow=Good	River Flow=Poor
Yes	Below Average		
Yes	Average		?
Yes	Above Average		
No	Below Average		
No	Average		
No	Above Average		

Rainfall	TreeCond	PesticideUse	PesticideInRiver	RiverFlow	Drought	FishAbundance
Average	Good	Yes	High	Good	No	Low
Average	Good	No	Low	Poor	No	High
Average	Damaged	Yes	High	Poor	Yes	Low
Below_average	Damaged	Yes	Low	Poor	Yes	Low
Below_average	Good	Yes	Low	Good	No	High
*	Damaged	Yes	Low	Poor	Yes	High
Below_average	Good	Yes	Low	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	No	Low	Good	*	High
Average	Good	Yes	High	Good	No	*
Above_average	Damaged	Yes	High	Good	No	Low
*	Good	Yes	Low	Good	Yes	High
Average	Dead	*	High	Poor	No	Low
Average	*	Yes	Low	Poor	No	Low
Average	Good	Yes	High	Poor	No	Low
Average	*	Yes	High	Good	No	Low
Average	Damaged	Yes	High	Good	No	Low
Average	Damaged	Yes	High	Good	No	*
Average	Damaged	*	High	Poor	No	High
Above_average	Dead	Yes	High	Good	Yes	*
Average	Good	Yes	High	Poor	No	High
Above_average	Good	Yes	Low	Good	No	High
Above_average	*	Yes	High	Good	No	High
Average	Good	Yes	High	Good	No	Low
Average	Good	Yes	High	Good	No	High
*	Good	Yes	High	Poor	No	*
Average	Dead	Yes	High	Poor	Yes	Low
Average	Damaged	Yes	Low	Good	No	High
Average	Damaged	Yes	High	Poor	Yes	Low
Average	*	Yes	Low	*	No	High
Below_average	Damaged	Yes	Low	Poor	Yes	Low
Average	Good	Yes	High	Poor	No	Low



Expectation Maximisation

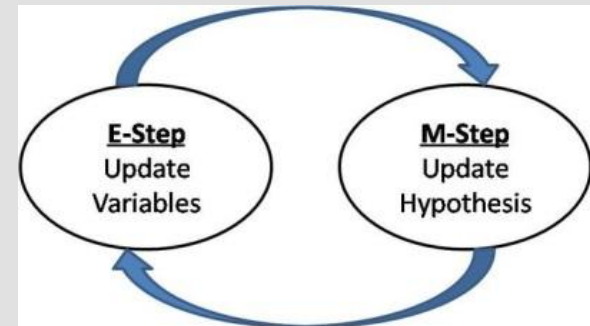
Iterative method is used to estimate missing data

Cycle between:

1. Estimating missing data
2. Learning parameters

Stop when no improvement in log likelihood

$$\log(p(\text{data}|\text{model}))$$



Learning Probabilities from Data

Netica Demo
(See lecture video)

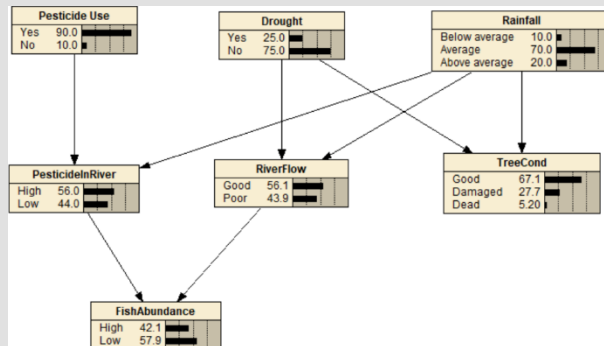


Evaluating BNs

- **With experts**
 - Face validity
 - Scenario testing
- **Evaluation against data**
 - Predictive accuracy, confusion matrices
 - Scoring functions (e.g. log score, Brier score)
- **Sensitivity analysis**
 - Sensitivity to findings or parameters

Evaluation against Data

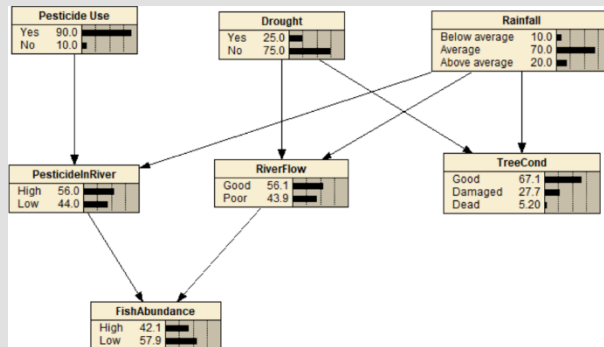
Use model to
generate a prediction
for each data case
and compare with the
actual value



PesticideUse	Drought	Rainfall	PesticideInRiver	RiverFlow	TreeCond	FishAbundance
*	*	Above_average	Low	*	Good	Low
Yes	No	Average	*	Good	*	Low
No	No	*	*	Good	Good	High
*	*	Average	Low	Good	*	High
*	Yes	*	High	*	*	Low
Yes	*	Average	Low	Good	Damaged	High
Yes	No	Average	*	Good	*	High
*	*	*	Low	*	*	High
*	*	*	High	*	*	High
*	*	*	Low	*	Good	High
Yes	*	*	High	*	*	Low
*	No	Average	*	Good	*	Low
Yes	No	Average	*	*	Good	High
*	No	Average	Low	*	Good	High
*	*	*	*	*	Good	Low
No	No	Average	Low	*	*	Low
Yes	*	*	*	*	*	Low
*	*	*	*	Poor	*	Low
Yes	*	Above_average	High	Good	*	Low
Yes	No	Average	High	*	*	High
*	No	*	*	*	*	High
*	*	*	High	Poor	Dead	High
Yes	No	*	High	Poor	Good	Low
No	No	*	*	*	Good	High
*	*	Average	*	*	Good	Low
Yes	*	*	Low	Good	*	High
Yes	No	*	*	Poor	*	Low
*	*	Above_average	High	Good	Damaged	Low
Yes	No	Average	Low	*	*	Low
Yes	*	Above_average	High	*	Good	Low
*	No	*	*	*	Good	Low
No	No	Above_average	Low	*	Good	High
Yes	No	Average	*	Good	Good	Low
Yes	*	*	High	Good	Good	Low
*	*	*	*	Poor	Good	High
*	Yes	Average	*	*	*	High
*	No	Below_average	Low	*	*	High
*	No	*	High	*	*	High
*	*	*	*	*	Damaged	High
*	*	Above_average	Low	*	*	High

Evaluation against Data

Use model to generate a prediction for each data case and compare with the actual value



PesticideUse	Drought	Rainfall	PesticideInRiver	RiverFlow	TreeCond	FishAbundance	P(FishAbundance=High)	P(FishAbundance=Low)
*	*	Above_average	Low	*	Good	Low	0.88695	0.11305
Yes	No	Average	*	Good	*	Low	0.522	0.478
No	No	*	*	Good	Good	High	0.828274	0.171726
*	*	Average	Low	Good	*	High	0.9	0.1
*	Yes	*	High	*	*	Low	0.177641	0.822359
Yes	*	Average	Low	Good	Damaged	High	0.9	0.1
Yes	No	Average	*	Good	*	High	0.522	0.478
*	*	*	Low	*	*	High	0.681049	0.318951
*	*	*	High	*	*	High	0.21579	0.78421
*	*	*	Low	*	Good	High	0.714233	0.285767
Yes	*	*	High	*	*	Low	0.215762	0.784238
*	No	Average	*	Good	*	Low	0.5535	0.4465
Yes	No	Average	*	*	Good	High	0.4176	0.5824
*	No	Average	Low	*	Good	High	0.72	0.28
*	*	*	*	*	Good	Low	0.43666	0.56334
No	No	Average	Low	*	*	Low	0.72	0.28
Yes	*	*	*	*	*	Low	0.395349	0.604651
*	*	*	*	Poor	*	Low	0.288789	0.711211
Yes	*	Above_average	High	Good	*	Low	0.27	0.73
Yes	No	Average	High	*	*	High	0.216	0.784
*	No	*	*	*	*	High	0.447786	0.552214
*	*	*	High	Poor	Dead	High	0.135	0.865
Yes	No	*	High	Poor	Good	Low	0.135	0.865
No	No	*	*	*	Good	High	0.693577	0.306423
*	*	Average	*	*	Good	Low	0.431051	0.568949
Yes	*	*	Low	Good	*	High	0.9	0.1
Yes	No	*	*	Poor	*	Low	0.277211	0.722789
*	*	Above_average	High	Good	Damaged	Low	0.27	0.73
Yes	No	Average	Low	*	*	Low	0.72	0.28
Yes	*	Above_average	High	*	Good	Low	0.266085	0.733915
*	No	*	*	*	Good	Low	0.446712	0.553288
No	No	Above_average	Low	*	Good	High	0.8955	0.1045
Yes	No	Average	*	Good	Good	Low	0.522	0.478
Yes	*	*	High	Good	Good	Low	0.27	0.73
*	*	*	*	Poor	Good	High	0.288495	0.711505
*	Yes	Average	*	*	*	High	0.318262	0.681737
*	No	Below_average	Low	*	*	High	0.63	0.37
*	No	*	High	*	*	High	0.228506	0.771494
*	*	*	*	*	Damaged	High	0.390934	0.609066
*	*	Above_average	Low	*	*	High	0.874125	0.125875

Confusion Matrix

Using the most likely state: $p > 0.5$ for binary nodes

PesticideUse	Drought	Rainfall	PesticideInRiver	RiverFlow	TreeCond	FishAbundance	Pred FishAbundance
*	*	Above_average	Low	*	Good	Low	High
Yes	No	Average	*	Good	*	Low	High
No	No	*	*	Good	Good	High	High
*	*	Average	Low	Good	*	High	High
*	Yes	*	High	*	*	Low	Low
Yes	*	Average	Low	Good	Damaged	High	High
Yes	No	Average	*	Good	*	High	High
*	*	*	Low	*	*	High	High
*	*	*	High	*	*	High	Low
*	*	*	Low	*	Good	High	High
Yes	*	*	High	*	*	Low	Low
*	No	Average	*	Good	*	Low	High
Yes	No	Average	*	*	Good	High	Low
*	No	Average	Low	*	Good	High	High
*	*	*	*	*	Good	Low	Low
No	No	Average	Low	*	*	Low	High
Yes	*	*	*	*	*	Low	Low
*	*	*	*	Poor	*	Low	Low
Yes	*	Above_average	High	Good	*	Low	Low
Yes	No	Average	High	*	*	High	Low
*	No	*	*	*	*	High	Low
*	*	*	High	Poor	Dead	High	Low
Yes	No	*	High	Poor	Good	Low	Low
No	No	*	*	*	Good	High	High
*	*	Average	*	*	Good	Low	Low
Yes	*	*	Low	Good	*	High	High
Yes	No	*	*	Poor	*	Low	Low
*	*	Above_average	High	Good	Damaged	Low	Low
Yes	No	Average	Low	*	*	Low	High
Yes	*	Above_average	High	*	Good	Low	Low
*	No	*	*	*	Good	Low	Low
No	No	Above_average	Low	*	Good	High	High
Yes	No	Average	*	Good	Good	Low	High
Yes	*	*	High	Good	Good	Low	Low
*	*	*	*	Poor	Good	High	Low
*	Yes	Average	*	*	*	High	Low
*	No	Below_average	Low	*	*	High	High
*	No	*	High	*	*	High	Low
*	*	*	*	*	Damaged	High	Low
*	*	Above_average	Low	*	*	High	High

Confusion Matrix

Using the most likely state: $p > 0.5$ for binary nodes

Actual	Predicted	
	High	Low
High	96	123
Low	38	234

Error rate = 32.8%

PesticideUse	Drought	Rainfall	PesticideInRiver	RiverFlow	TreeConc	FishAbundance	Pred FishAbundance
*	*	Above_average	Low	*	Good	Low	High
Yes	No	Average	*	Good	*	Low	High
No	No	*	*	Good	Good	High	High
*	*	Average	Low	Good	*	High	High
*	Yes	*	High	*	*	Low	Low
Yes	*	Average	Low	Good	Damaged	High	High
Yes	No	Average	*	Good	*	High	High
*	*	*	Low	*	*	High	High
*	*	*	High	*	*	High	Low
*	*	*	Low	*	Good	High	High
Yes	*	*	High	*	*	Low	Low
*	No	Average	*	Good	*	Low	High
Yes	No	Average	*	*	Good	High	Low
*	No	Average	Low	*	Good	High	High
*	*	*	*	*	Good	Low	Low
No	No	Average	Low	*	*	Low	High
Yes	*	*	*	*	*	Low	Low
*	*	*	*	Poor	*	Low	Low
Yes	*	Above_average	High	Good	*	Low	Low
Yes	No	Average	High	*	*	High	Low
*	No	*	*	*	*	High	Low
*	*	*	High	Poor	Dead	High	Low
Yes	No	*	High	Poor	Good	Low	Low
No	No	*	*	*	Good	High	High
*	*	Average	*	*	Good	Low	Low
Yes	*	*	Low	Good	*	High	High
Yes	No	*	*	Poor	*	Low	Low
*	*	Above_average	High	Good	Damaged	Low	Low
Yes	No	Average	Low	*	*	Low	High
Yes	*	Above_average	High	*	Good	Low	Low
*	No	*	*	*	Good	Low	Low
No	No	Above_average	Low	*	Good	High	High
Yes	No	Average	*	Good	Good	Low	High
Yes	*	*	High	Good	Good	Low	Low
*	*	*	*	Poor	Good	High	Low
*	Yes	Average	*	*	*	High	Low
*	No	Below_average	Low	*	*	High	High
*	No	*	High	*	*	High	Low
*	*	*	*	*	Damaged	High	Low
*	*	Above_average	Low	*	*	High	High

Netica also computes ROC curves, which show the effect of varying the cutoff threshold

Scoring Rules – Log Score

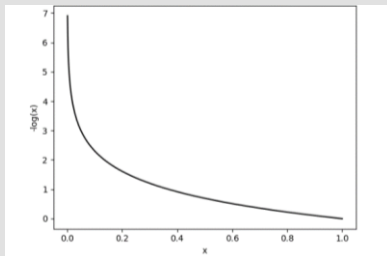
Orthodox measures ignore the uncertainty of the prediction

Instead, we can score each prediction using a log score:

$$-\log P(\text{true class})$$

Range between zero and infinity

Predicts true class is impossible (i.e., 0%) = Infinity



Predicts true class with 100% confidence = Zero

PesticideUse	Drought	Rainfall	PesticideInRiver	RiverFlow	TreeCond	FishAbundance	P(True Class)	-Log(P(True Class))
*	*	Above_average	Low	*	Good	Low	0.1131	2.1799
Yes	No	Average	*	Good	*	Low	0.4780	0.7381
No	No	*	*	Good	Good	High	0.8283	0.1884
*	*	Average	Low	Good	*	High	0.9000	0.1054
*	Yes	*	High	*	*	Low	0.8224	0.1956
Yes	*	Average	Low	Good	Damaged	High	0.9000	0.1054
Yes	No	Average	*	Good	*	High	0.5220	0.6501
*	*	*	Low	*	*	High	0.6810	0.3841
*	*	*	High	*	*	High	0.2158	1.5334
*	*	*	Low	*	Good	High	0.7142	0.3365
Yes	*	*	High	*	*	Low	0.7842	0.2430
*	No	Average	*	Good	*	Low	0.4465	0.8063
Yes	No	Average	*	*	Good	High	0.4176	0.8732
*	No	Average	Low	*	Good	High	0.7200	0.3285
*	*	*	*	*	Good	Low	0.5633	0.5739
No	No	Average	Low	*	*	Low	0.2800	1.2730
Yes	*	*	*	*	*	Low	0.6047	0.5031
*	*	*	*	Poor	*	Low	0.7112	0.3408
Yes	*	Above_average	High	Good	*	Low	0.7300	0.3147
Yes	No	Average	High	*	*	High	0.2160	1.5325
*	No	*	*	*	*	High	0.4478	0.8034
*	*	*	High	Poor	Dead	High	0.1350	2.0025
Yes	No	*	High	Poor	Good	Low	0.8650	0.1450
No	No	*	*	*	Good	High	0.6936	0.3659
*	*	Average	*	*	Good	Low	0.5689	0.5640
Yes	*	*	Low	Good	*	High	0.9000	0.1054
Yes	No	*	*	Poor	*	Low	0.7228	0.3246
*	*	Above_average	High	Good	Damaged	Low	0.7300	0.3147
Yes	No	Average	Low	*	*	Low	0.2800	1.2730
Yes	*	Above_average	High	*	Good	Low	0.7339	0.3094
*	No	*	*	*	Good	Low	0.5533	0.5919
No	No	Above_average	Low	*	Good	High	0.8955	0.1104
Yes	No	Average	*	Good	Good	Low	0.4780	0.7381
Yes	*	*	High	Good	Good	Low	0.7300	0.3147
*	*	*	*	Poor	Good	High	0.2885	1.2431
*	Yes	Average	*	*	*	High	0.3183	1.1449
*	No	Below_average	Low	*	*	High	0.6300	0.4620
*	No	*	High	*	*	High	0.2285	1.4762
*	*	*	*	*	Damaged	High	0.3909	0.9392
*	*	Above_average	Low	*	*	High	0.8741	0.1345

Netica also computes similar scoring rules: Quadratic Loss and Spherical Payoff



Evaluation against Data

**Netica Demo
(See lecture video)**



Sensitivity analysis

- **Sensitivity to findings**

- How would observing evidence affect the probability of your query node?
- Graphical approaches, e.g., d-separation
- Approaches that use uncertainty metrics

- **Sensitivity to parameters**

- How would changes to model parameters change the probability of a query node or model behaviour

Sensitivity analysis – uncertainty metrics

- **Uncertainty metrics in Netica:**

- Mutual Information, $I(X, Y)$

$$I(X, Y) = \sum_x \sum_y \Pr(X = x, Y = y) \log\left(\frac{\Pr(X = x, Y = y)}{\Pr(X = x) \Pr(Y = y)}\right)$$

- Variance of beliefs

$$\sum_x \sum_y \Pr(X = x, Y = y) (\Pr(X = x|Y = y) - \Pr(X = x))^2$$



Properties of Mutual Information

- $I(X, X) = \text{Entropy of } X = H(X)$

$$H(X) = - \sum_x \text{Pr}(x) \log(\text{Pr}(x))$$

- $I(X, Y) = H(X) + H(Y) - H(X, Y)$
- $I(X, Y) \geq 0$ (*i.e. positive*)
- $I(X, Y) = I(Y, X)$ (*i.e. symmetric*)
- $I(X, Y) = 0$ ***iff X and Y are independent.***

Sensitivity to findings

**Netica Demo
(See lecture video)**





FIT3080 – Artificial Intelligence

Decision Networks

Russell & Norvig Ch 16.1-2, 16.5

Korb & Nicholson, 2011: Ch 4.1-4.4

How do people make decisions?

Decision making under uncertainty is ubiquitous.

- **Shall I take my raincoat today?**
 - Will it rain?
- **Shall I try the prawns?**
 - Will they be safe?
 - Would I prefer them to steak?

Bayesian decisions

Decision problems have two aspects:

- **Probability:**
What is the probability I will catch COVID this month?
- **Utility:**
How bad would it be if I do catch COVID?

Utility measures preferences

- Do you prefer having COVID to not having COVID?
- By how much?

Utility is also known as 'value' or (in some places) 'consequence'



Decision Networks

- Decision theory =
Utility theory + Probability theory
- Decision networks =
Utilities & actions + Bayesian networks

Also called *Influence Diagrams* (Howard & Matheson 1981)

Bayesian Decision Theory

- Frank Ramsey (1926)
- Decision making under uncertainty – what action to take when the state of the world is unknown
- Bayesian answer –
Find the utility of each possible outcome (action-state pair), and take the action that maximizes the *expected utility*

Expected Utility

$$EU(A | E) = \sum_i \Pr(O_i | E, A)U(O_i | A)$$

- E = available evidence
- A = a non-deterministic action
- O_i = a possible outcome state
- U = utility



Bayesian Decision Theory – Example

Action	Rain (p=0.4)	Shine (1-p=0.6)
Take umbrella	30	-10
Leave umbrella	-100	50

$$EU(A | E) = \sum_i \Pr(O_i | E, A) U(O_i | A)$$

Expected utilities:

$$\square E(\text{Take umbrella}) = 30 \times 0.4 + (-10) \times 0.6 = 6$$

$$\square E(\text{Leave umbrella}) = -100 \times 0.4 + 50 \times 0.6 = -10$$



Bayesian decision theory - example

$$EU(A | E) = \sum_i \Pr(O_i | E, A)U(O_i | A)$$

Example (Global Warming)

Action	GWH(p)	¬GWH (1-p)
Do Nothing	u_1	u_2
Curtail CO ₂	u_3	u_4

Expected utilities:

$$EU(\text{Do Nothing}) = p \times u_1 + (1 - p) \times u_2$$

$$EU(\text{Curtail CO}_2) = p \times u_3 + (1 - p) \times u_4$$

When people disagree on Global Warming, they may be disagreeing on p (probability that it is true) or on one or more of u_1 , u_2 , u_3 , u_4 .

Sensitivity – Example (Rain again)

Action	Rain (p=0.4)	Shine (1-p=0.6)
Take umbrella	30	-10
Leave umbrella	-100	50

$$EU(A | E) = \sum_i \Pr(O_i | E, A) U(O_i | A)$$

Expected utilities:

$$\square E(\text{Take umbrella}) = 30 \times 0.4 + (-10) \times 0.6 = 6$$

$$\square E(\text{Leave umbrella}) = -100 \times 0.4 + 50 \times 0.6 = -10$$



Sensitivity – Example (Rain again)

Action	Rain (p=0.3)	Shine (1-p=0.7)
Take umbrella	30	-10
Leave umbrella	-100	50

$$EU(A | E) = \sum_i \Pr(O_i | E, A) U(O_i | A)$$

Expected utilities:

□ $E(\text{Take umbrella}) = 30 \times 0.3 + (-10) \times 0.7 = 2$

□ $E(\text{Leave umbrella}) = -100 \times 0.3 + 50 \times 0.7 = 5$

!!



Indifference

$EU(A_1) = EU(A_2)$ given some probability distribution

e.g. $EU(\text{Take umbrella}) = EU(\text{Leave umbrella})$

$$30p + (-10)(1 - p) = -100p + 50(1 - p)$$

$$130p = 60 - 60p$$

$$190p = 60$$

$$p = \frac{6}{19} \approx 0.32$$

$EU(\text{Take umbrella}) < EU(\text{Leave umbrella})$ when $p < \sim 0.32$

Strategy: Take umbrella if $P(\text{rain}) > 0.32$, otherwise leave it

→ Can use this in reverse (i.e. fix p) to work out a person's utilities

Fever Problem Description

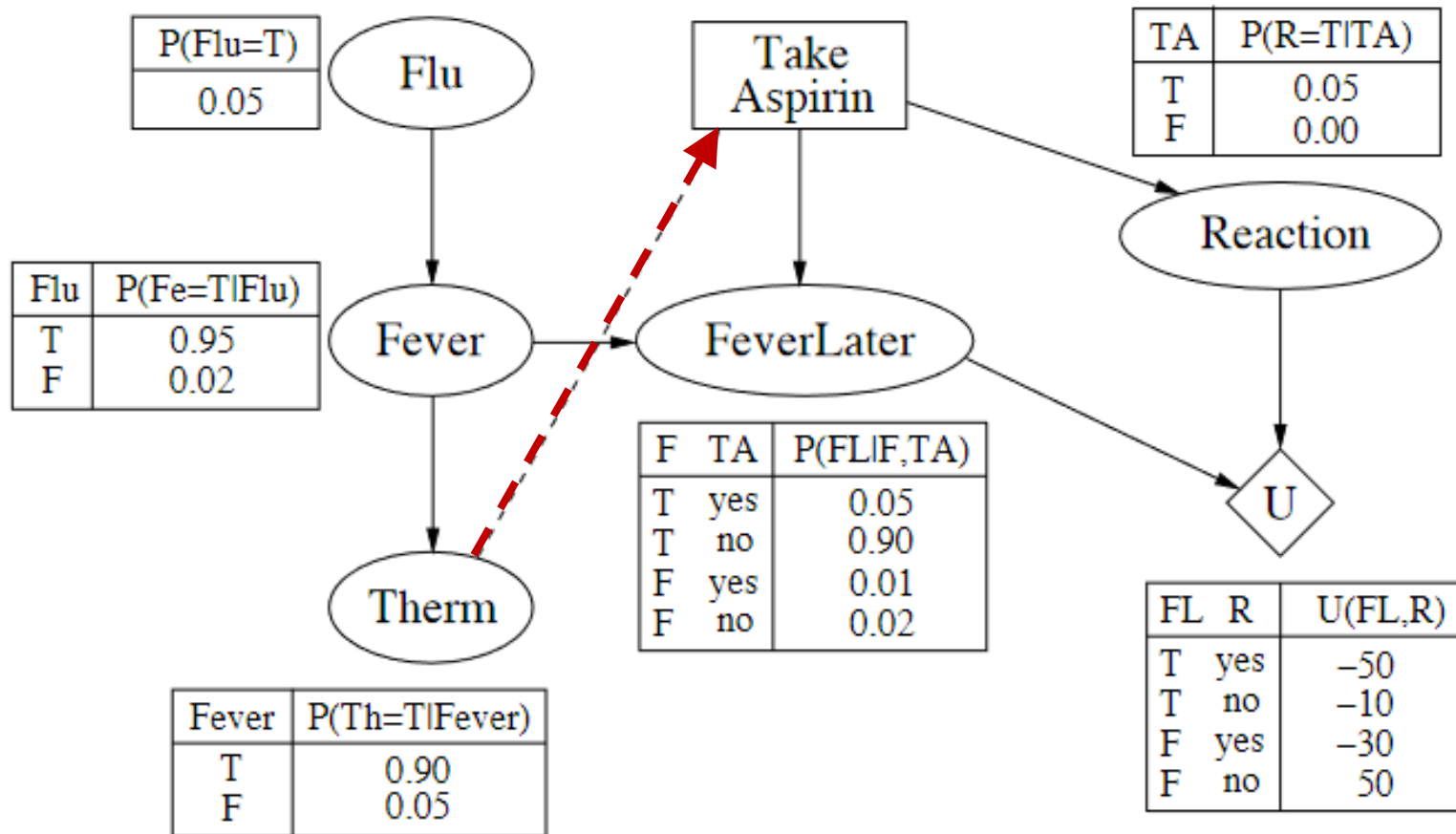
Suppose you know:

- **Fever can be caused by flu**
- **A thermometer is fairly reliable at indicating fever**
- **Taking aspirin almost certainly lowers a fever to normal**
- **Some people (about 5% of the population) have a negative reaction to aspirin**

You'll be happy to get rid of your fever, so long as you don't suffer an adverse reaction if you take aspirin.



Fever Decision Network



K&N Figure 4.5

Types of Nodes

- **Chance nodes – (ovals) random variables**
 - Have an associated CPT
 - Parents can be decision nodes and other chance nodes
- **Decision nodes – (rectangles) points where the decision maker has a choice of actions**
 - The table is the decision with the highest computed EU for each combination of evidence in the *information link* parents
- **Utility nodes (Value nodes) – (diamonds) the agent's utility function**
 - The table represents a multi-attribute utility function
 - Parents are variables describing the outcome states that directly affect utility

Types of Links

- **Information Links** – indicate when a chance node needs to be observed before a decision is made
 - Any link entering a decision node is an informational link
- **Precedence links** – indicate the order in which decisions are made

Fever Decision Network

**Netica Demo
(See lecture video)**

Bayesian networks – applications

- **Decision making under uncertainty**
 - Risk assessment
 - Decision support in high uncertainty/high consequence settings (e.g., in medicine and law)
- **Causal inference**
 - Modelling a data generating process to isolate real causal effects from correlations
 - Increasingly common in epidemiology, and other areas of statistics-based science
- **Long term forecasting**
 - Projecting potential outcomes where there is little data, but good expert knowledge
- **Complex system modelling**
 - Common for modelling complex ecological and environmental systems
 - Frequently uses extensions such as dynamic, spatial and object-oriented Bayesian networks

Next Lecture Topic

- **Lecture Topic 6**
 - Markov Decision Processes (MDPs)

